## A real consultation

PhysicianBench · hepatology e-consult

**Lab alert in the inbox**

Abnormal iron studies and liver tests in a patient with known fatty liver disease.

**The agent must**

1. Review the patient's record
2. Interpret the iron and liver labs
3. Order the diagnostic workup
4. Write the assessment note

**Real de-identified patient record**

FHIR API · search, order and note tools

**Graded by 7 checkpoints**

data retrieval · ferritin · Gilbert · lifestyle · HFE gene test (order) · FibroScan (order) · assessment note

retrieval · reasoning · action · documentation

## The agent at work

one decision per step, then the tools run

1. patient
2. problems · labs · meds · social
3. labs
4. labs ×4
5. labs
6. labs ×6
7. notes · past orders
8. **decision point: which turn next?**
9. write note
10. finish

The executed run passes 7 / 7 checkpoints.

## Step 8: same state, five proposals

the LLM proposes K = 5 turns; who picks one?

| | Proposals (sampled from the LLM) | System One: p(option) |
|---|---|---|
| LLM | A Search labs ×3 · vitals | 0.05 |
| | B Search clinical notes | 0.16 |
| | C Search notes · labs ×2 | 0.09 |
| | D Search clinical notes | 0.23 |
| S1 | E **Place 6 orders** incl. HFE gene test and FibroScan | **0.47** |

**LLM decides**

samples one turn: A

**Keeps searching; no order at this step.**

Across 42 such steps its own sample orders in 38%.

**S1 System One decides**

one pass over all five: E

**Places both required orders**

HFE gene test, FibroScan

Across 42 such steps it places the order in 100%.

Same task · same record · same tools · same proposals · same graders — only the decision maker changes.